

Classwork 3

Quadratics and the discriminant

Grades: [A] [M] [E]

Show all working. Give answers to 3 decimal places unless the question asks for an exact answer.

Part A · Solve, count, build

A1 Solve $(2x - 1)(x + 4) = 0$. [A]

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A2 Solve $x^2 - 3x = 28$. [A]

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A3 Solve $3x^2 + 2x - 4 = 0$. [A]

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A4 Find the discriminant of $x^2 - 4x + 7 = 0$ and say what it tells you. [A]

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A5 Write a quadratic equation with roots $\frac{2}{3}$ and -1 , with whole-number coefficients. [A]

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Part B · Find the unknown

B1 $3x^2 + kx + 12 = 0$ has equal roots. Find k . [M]

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B2 $x^2 + 8x + p = 0$ has two different real roots. Find the values of p . [M]

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B3 A parabola crosses the x -axis at -3 and 2 and passes through $(1, -8)$. Find its equation. [M]

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Part C · Real roots for every k

C Show that $x^2 - (k + 3)x + 3k = 0$ has real roots for every value of k , and find when the roots are equal. **[E]**

Words you may use

discriminant · perfect square · never negative · therefore · equal roots

Sentence starters (optional): *The discriminant is ... This is a perfect square, so ... The roots are equal when*

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Extension

Solve $x^4 - 5x^2 + 4 = 0$.

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